

An Exact Event Atlas for a Two-Threshold Hysteretic Relay Oscillator

Route-A source-local certificate HCS-C252

30 August 2026

Abstract

We give a complete, guard-specific analysis of a two-threshold relay phase oscillator with a transverse linear cocycle. The phase legs, Poincaré map, periodic set, grazing convention, and no-Zeno bound are all exact for every $h > 0$ and $\gamma \geq 0$. Eight rational receipts are independently checked, symbolically replayed, and hostile-tested. This is a source-local hybrid theorem: no arithmetic origin, Euler factor, root number, or Hilbert–Polya operator is claimed.

1 Frozen hybrid model

The state is (θ, y, σ) with $-h \leq \theta \leq h$, $y \in \mathbb{R}$, and $\sigma \in \{-1, +1\}$. In the interior,

$$\dot{\theta} = \sigma, \quad \dot{y} = -\gamma y, \quad h > 0, \gamma \geq 0. \quad (1)$$

The guard has priority at equality: $\theta = h$ with $\sigma = +1$ is reset to $\sigma = -1$, and $\theta = -h$ with $\sigma = -1$ is reset to $\sigma = +1$. There is no sliding state in this frozen convention. The return section is $\Sigma_- = \{(\theta, y, \sigma) = (-h, y, +1)\}$.

2 Event theorem

Theorem 1 (exact relay atlas). *Every consistent interior or boundary state has a unique forward execution. A leg from one threshold to the other has duration $2h$ and maps $y \mapsto e^{-2\gamma h}y$. Therefore the Poincaré map on Σ_- is*

$$P(y) = e^{-4\gamma h}y, \quad T_{\text{phase}} = 4h. \quad (2)$$

For $\gamma > 0$, the unique periodic state in the transverse coordinate is $y = 0$; for $\gamma = 0$, every y is periodic. Every interior point reaches a switching section in at most $2h$, and all successive event gaps equal $2h$.

Proof. On each mode, $\theta(t) = \theta_0 + \sigma t$ reaches the opposite threshold in time $2h$; the equality rule chooses the only outgoing mode. Solving the scalar equation in (1) gives $y(t) = y_0 e^{-\gamma t}$, so composition gives (2). Since $h > 0$, the event gaps have a uniform positive lower bound, which rules out finite-time Zeno accumulation. The fixed-point equation for (2) gives the stated periodic set. $\square$

3 Boundary and stability atlas

The multiplier of the full return is $m = e^{-4\gamma h}$. Thus $0 < m < 1$ for $\gamma > 0$, while $m = 1$ on the neutral face $\gamma = 0$. The zero section $y = 0$ is an invariant periodic phase orbit and is called the grazing label only to identify the transverse fixed set; it does not create a sliding segment. At $h = 0$ the event lower bound disappears, so the face is excluded rather than regularized. An inconsistent boundary mode is corrected instantaneously by the guard before flow starts.

Table 1: Exact receipt coverage (all inputs are rational).

regime	rows	certified quantities
contracting $\gamma > 0$	5	leg map, return, multiplier, period
neutral $\gamma = 0$	1	continuum of fixed transverse levels
zero transverse amplitude	1	invariant periodic phase
grazing-labelled boundary	1	guard policy and no sliding

4 Executable certificate

The producer emits eight rows and two leg records per row. The producer- independent checker closes 189 assertions; a separate SymPy program closes ten flow, composition, guard, and semigroup identities. Two fresh producer runs are byte identical and the hostile suite rejects 21/21 altered payloads. Decimal exponentials are only serialized evaluations of the exact semigroup. The three manuscript rounds use fixed `SOURCE_DATE_EPOCH=1788048000`, embedded fonts, text extraction, and a content-addressed manifest.

5 Route-A boundary

The strict evaluator tuple is `(AO_FAIL,A1_PASS_ANALYTIC,A2_FAIL,A3_FAIL,A4_FORMAL_HINT)`. The arithmetic origin is `none`; no primitive target orbit, determinant, or target data are defined. The verdict is `ROUTE_A_REJECTED`, with `route_b_invocation_allowed: false`, under the scope literal `NO_BAD_EULER_OR_ROOT_NUMBER`. This event map is a hybrid mechanical receipt, not a target spectral determinant.

6 Conclusion

A precise hysteresis convention turns a relay intuition into an auditable global event theorem. The exact return multiplier and the uniform event gap separate transverse stability from geometric period and make the stopping boundary transparent. No arithmetic bridge is inferred.

References

- [1] D. Liberzon, *Switching in Systems and Control*, Birkhäuser, 2003.
- [2] A. F. Filippov, *Differential Equations with Discontinuous Righthand Sides*, Kluwer, 1988.
- [3] M. di Bernardo et al., *Piecewise-smooth Dynamical Systems*, Springer, 2008.